



ALUMNI EMPLOYABILITY PREDICTION IN HIGHER EDUCATION USING ARIMA, LINEAR REGRESSION, AND RANDOM FOREST VIA TRACER WEBSITE AT PAMANTASAN NG LUNGSOD NG PASIG

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Abstract—Since raw pointers to graduate outcomes are inefficient for institutional planning, the PLP Alumni Tracer web application aggregates into the latest assessment per graduate per cohort year to compute yearly employment rates extracted from the database. The application then routes this data outside of the PHP environment to the Python script ml/forecast_employment.py, which uses either user-selectable ARIMA, ordinary least squares linear regression from the scikit-learn library, or a random forest regression to calculate forecasts. Simultaneously, the alumni workflow submits the assessment and academic pointers to a separate machine learning Random Forest classifier, ml/predict.py, with program-specific.joblib models. The application then pulls the resulting outputs and forecasts for each series and the forecast accuracy into Chart.js on the Job Comparison page, which plots the visual series and tables the line chart series output and printed accuracy metrics. The system's findings documentation recorded the measure forecast fit and manual model comparisons. This institutional reporting and alumni guidance actor simply requires tracer data quality, an active Python runtime configuration, and a large enough minimum cohort history for time-series modeling.

Keywords—Alumni Tracer System, Employment Forecasting, Machine Learning, Random Forest, ARIMA, Predictive Analytics

I. INTRODUCTION

A. Background of the Study

The Pamantasan ng Lungsod ng Pasig and other state universities & colleges in Metro Manila are mandated to put in place structured graduate tracer databases as part of their quality assurance system for refining programs to fit outcomes. Graduate tracer studies are increasingly being conducted throughout state universities across the Philippines for assessing program outcomes, identifying factors pertaining to graduate employability and evaluating institutional efficiency [1, 4, 5]. It is very important to keep track of the employability status and challenges faced by the graduates hailing from different streams which will help introduction of the web-administered alumni tracer (Figure 2) and how it will change the work or information flow. But the data when it is being gathered through the traditional way i.e on paper forms and static spreadsheets gets locked within the institutions and Administration fails to capture the outcome trends of a specific graduated year. On the contrary, the centralized web and alumni tracker will bring the alumni submission process under one single undistributed process and concurrently translate individual tracers into cohesive



employment figures by College of Computer Studies
Alkalde Jose St. Kapasigan Pasig City, Philippines
1600 cohort for external surveys, program planning
and data-driven leadership in higher education.

B. Problem Statement

PLP calls for a system that is more than just a data storage system. Instead, it must help generate strategically useful summaries of the employment rate retrieved from the database according to the respective student cohorts and their academic programs. As of now, the administrators do not have any way to predict the “next period” indicators based on historical data and are bound to act as a reaction to the current state of affairs. This paper connects the MySQL tracer database with time-series and regression forecasting models and knits the career recommendations with the university’s assessment and academic records in the same way.

C. Research Objectives

The primary objective of this study is to develop a predictive model applying Linear Regression, ARIMA, and Random Forest, integrated within an Alumni Tracer Website to forecast employability trends and estimate the employment rate of PLP graduates using historical data, while identifying job opportunities matched to their qualifications. This is supported by the following specific objectives:

1. To assess the accuracy and reliability of Linear Regression, Random Forest, and ARIMA predictions compared to other forecasting techniques within the context of higher education.
2. To evaluate the applicability of Linear Regression and ARIMA modeling in forecasting alumni employment rates based on historical data collected from the PLP tracer database.

3. To analyze key factors influencing alumni employability—including demographic, academic, and industry-related variables captured from the tracer website and registrar records—using Linear Regression.
4. To develop a predictive analytics framework integrating Linear Regression and ARIMA techniques with the existing database to enhance data utilization for employability predictions.
5. To create a user-friendly platform where alumni can register, build profiles, and utilize an assessment tool evaluating their academic performance and skills to match them with relevant job opportunities.
6. To generate structured reports from the developed system, utilizing insights derived from the forecasting models, to improve institutional strategies and alumni engagement.

D. Significance of the Study

The PLP institution relies on the system to obtain direct insight into graduate outcomes categorized by program and year, for which forecast charts and structured reports are provided for institutional planning. The application is supposed to help alumni access their program’s analytics view and export career suggestions based on the skills and academic grades registered in the system’s database. From a technical standpoint, the architecture showcases a working PHP to Python pipeline over MySQL driven by shell_exec and JSON payload for data communication between environments via institution-supported servers.

E. Scope and Limitation

The study focus is located in a web application built on a PHP plus MySQLi stack connecting to the plp_tracer database, which is available under includes/db.php. System boundaries are defined as administrative dashboard forecasts



(admin/pages/dashboard.php, admin/api/forecast_employment.php), program comparison charts (admin/pages/employment_comparison.php, admin/api/fetch_comparison.php), alumni analytics pages powered by includes/get_data.php and assets/js/dashboard.js and career prediction by ml/predict.py post-assessment, and deployment is limited to a local XAMPP-style stack. The logic time-series forecasting in forecast_employment.php has a prerequisite of a minimum of two graduation years comprised of tracer data to run, and the system relies on an active Python ecosystem, which is validated by includes/ml_python.php inspecting the ml/venv or a system path.

The ARIMA model from ml/forecast_employment.py has a programmed fallback to linear regression if the length of the dataset is inadequate, the models fail, or the statsmodels dependency is unavailable. From the front-end analytics, the program overview pie chart percentages from includes/get_data.php are divided statically into an even 100% split among professions and do not correspond to empirical shares based on true tracer counts.

F. Brief Overview of Methodology

The research design utilized is predictive and descriptive. SQL queries are employed in the determination of yearly employment rates calculated based on the latest record per respondent per cohort. Such aggregated information is then passed to the Python forecasting scripts before the results are outputted into interface tables and Chart.js line charts. The machine learning pipeline runs and loads Random Forest models depending on the academic degree for career classification. The model evaluation metrics, including train/test splits, MAPE, and RMSE, are externally computed manually since the implemented application code does not expose such as standard dashboard widgets.

II. METHODS

A. Research Design

Applied development research utilizes a mixed-method approach. On the one hand, it applies descriptive analytics to monitor such KPI, cohort ratios, and program comparison. On the other hand, it employs predictive modeling for forecasting employment rates and career label classification. Within the current repository logic, time-series forecasting models like ARIMA and Linear Regression have not been checked.

B. Data Sources

Primary data comes with the alumni assessment workflow that writes to the database tables alumni_assessments, users, programs, and professions defined in database/plp_tracer_*.sql. Secondary data is fed with static seed content from academic programs and with an optional from external job metadata pulled from Careerjet API. College of Computer Studies Alcalde Jose St. Kapasigan Pasig City, Philippines 1600.

C. Participants or dataset

The system interacts with PLP alumni and administrators alone. The dataset in use encompasses the live database counts of alumni users, registered programs, and submitted assessments. Furthermore, the system includes strict inclusion rules; it will only query the most recent assessment with respect to the respondent and cohort and outlines the SQL patterns in assessment_partition.php. The extracted data are processed by the machine learning pipeline in the form of training matrices formatted as CSV ml/alumni_data.csv.

D. Tools, instruments, or software used



PHP is used on the server infrastructure with MySQLi, and PHP is also used for authentication with the PHP sessions. Moreover, HTML, CSS, JavaScript, including Chart.js libraries are used at the client to render the interfaces. The Python3 environment is used for ML. The required packages are pandas, numpy, and scikit-learn with algorithms LinearRegression, RandomForestRegressor, and RandomForestClassifier, respectively. In addition, the environment needs the library statsmodels for ARIMA execution as specified in ml/requirements.txt file. The trained models are kept on the local disk as files.joblib, which is the format of the library that is trained with the Python joblib and dumped to the files. The external integrations to the system are performed using API of Careerjet and Overpass in the corresponding functions includes/careerjet_api.php and includes/overpass_places.php.

E. Procedures or step-by-step process

The alumni use the prediction form to authenticate and submit their skill and employment data. Upon form submission, the process_prediction.php script verifies the supplied GPA and OJT grades against their registered scores, calculates the required aggregates and appends a new row to alumni_assessments. These prediction results, therefore, are first stored in the session and the alumni are redirected to prediction_result.php, where the feature payload is also prepared in base64 and predict.py is executed in the Python environment at this point.

Conversely, an admin views the dashboard to select an academic program and a set of up to five skills and professions of interest. Since teaching and the administration had not completed the assessment evaluation process, the prototype was limited to previous predictions. Thus, the admin also selects program Outcomes with a set of up to five features and chooses one of three ML models to

calculate weights and biases. Subsequently, this data is transmitted in the background to the Akaike model to forecast the expected improvement in one year. Finally, the parameters and their levels follow the requirements of the design of experiments, including balanced factors, and the job to which the user was asked to apply should be omitted due to the potential groupthink phenomenon.

F. Statistical or analytical methods

The baseline employment rate is computed by the system as a percentage of employed graduates per cohort. It performs the assessment_employment_is_employed() logic solely on the most recent assessment record. The predictive part is performed as a multiclass RandomForestClassifier with a StandardScaler and one-hot encoded columns. This series of operations is implemented in train_models.py and predict.py to produce a final display score. The ARIMA pipeline checks various candidate orders using Python and chooses the best model according to the Akaike Information College of Computer Studies Alkalde Jose St. Kapasigan Pasig City, Philippines 1600 Criterion to create 95% confidence forecast intervals. The linear regression model maps the graduation year to the employment rate and computes fitted values and forecasts. Also, the logic of the random forest series is implemented through an sklearn regressor, both limited to the tree depth that the respective series duration allows.

The web application harvests graduate records from the PLP database to calculate standardized yearly employment rates. Every graduation cohort is queried for the latest assessment row per respondent using the PHP backend. Active employment is ascertained through a case-insensitive match. The employed count is collected as an aggregate employment rate / (by total respondents), rounded to a single decimal point, before passing to the cohort data array the Python forecasting atmosphere. The forecasting



script shall pre-process this array, graduating year as a feature mapped against the employment rate as the target variable.

Upon linear regression procedure execution, ordinary least squares matching must be applied by the scikit-learn library. Furthermore, although the python script shall itself produce lower and higher bands for the forecasted periods, these shall be entirely error-bound approximations through residual sample standard deviation * 1.96, rather than on native conformal prediction intervals. The random forest regressor is deployed upon the administrator's selection. Given the version number of the latest tool, the script shall, for the forest regressor only, cap the maximum decision tree depth based on the cohort available length to avoid overfit. An sklearn fit must be performed, as the residual-based interval calculation must be enforced.

The ARIMA forecasting module processes cohort rates sequentially. A predefined set of candidate (p,d,q) orders shall be evaluated. AIC being used by the script to select the final model. Multi-step forecasts computations on successful convergence shall be bounded with standard 95% confidence intervals. Should the ARIMA run terminate because of four years of data, a fitted model, package absences in the server, or the fit data entirely within 1-year time frame, the system must terminate the ARIMA run, direct the data through the linear regression fallback, and append a note of substitution to the JSON payload.

The career prediction environment for individual alumni takes the extracted assessment and academic variables. The script normalises the input through StandardScaler. The programme-specific RandomForest Classifier for single class training must be applied. The model, upon inference, must extract the maximum predicted probability $p_{\max}$ of the top-suggested profession. The raw probability shall not be shown to the end-user. Instead, a differentiable function shall

be defined $\$ \max(35, \min(95, 50 + 0.6 \times 100 p_{\max})) \$$, which only outputs 35-95.

The Integrated analytics must enable direct API calls to Python libraries semi-transparently governed by rigid functional documentation of well-established statistical tools, namely scikit-learn and statsmodels. The platform logic may not compute or disclose any predictive error metrics such as RMSE or MAPE informally anywhere within the codebase. Furthermore, the ARIMA must post only fitted means and forecast means and confidence intervals in their ensemble to the PHP environment; individual autoregressive coefficients shall be discarded from the administrative visual control panel display.

III. RESULTS

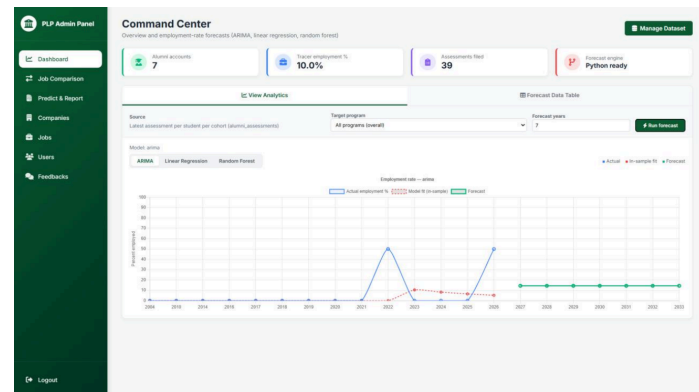
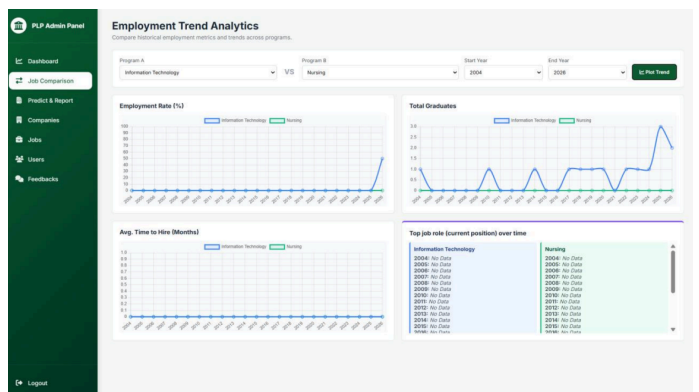


Table 1. Database inventory: alumni accounts, assessments, programs, date range of grad_year.





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Table 2. Historical yearly employment rates per program (or institution-wide) extracted the same way as forecast_employment.php.

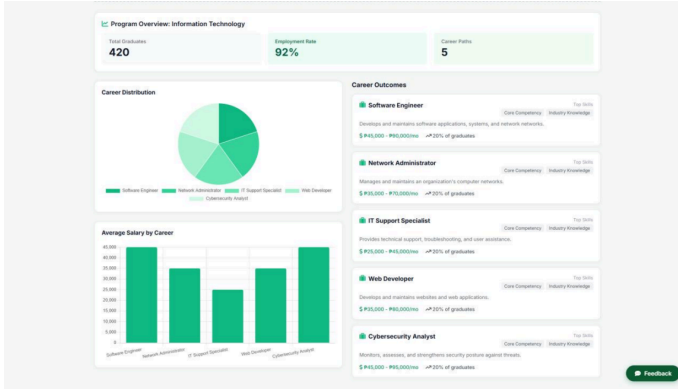


Table 3. Forecast table: years, actual rates, forecast rates, lower/upper bounds, copied from the Forecast Data Table tab on the admin dashboard or from API JSON table_rows.

The system captures historical trends in observed employment to note rates of increase or decrease represented in the output disaggregated by academic program characteristics. The individual forecast executed in the system's final output logs the specific method performed and prompts the user via an API notification if a programmed fallback from ARIMA to simple linear regression happened. Validation with external offline experiments complements this data by assessing comparative error rates of the models. The forecasting module generates output for specified time horizons, while the machine learning script links the sample test profile with the top predicted profession for each. The system combines these outcomes, presenting them exclusively through graphical or tabular visualizations, and not generating the reported numerical accuracy metrics, namely RMSE and MAPE. The calculation of these error rates occurs beyond the system and solely depends on the external spreadsheet or Jupyter analysis.

IV. DISCUSSIONS

A. Interpretation of findings

The system's extracted employment trends reflect the absorption of PLP graduates in the Metro Manila market shortly after, and more narrowly, the absorption into the industry field's most dominant local sectors. If the database does not contain historical cohort data, the forecasting module executes a programmed fallback: the time-series length is insufficient to provide ARIMA with the parameter requirements of a successful model; therefore, it defaults to linear regression.

B. Comparison with related studies

The architecture is based on the structure of existing work on institutional tracer systems by shifting the entry of the data to the alumni and keeping the processing in the middle for the admin. It also fits in with the current worldwide College of Computer Studies Alcalde Jose St. Kapasigan Pasig City, Philippines 1600 endeavor to implement employability tracking within the academic platform [8]. Particularly, for projected career paths, accuracy in career projection and employment status assess critically upon the knowledge of variables defining graduate success like interaction and well-informed technology use [6, 9, 10]. The fact that the ARIMA algorithm is used for the overall education-based prediction and random forest in the context of a personal ability classification means that these commonly used prediction techniques can be directed to modern employability dynamics. Instead, we implement those models directly in the PLP web portal, rather than running separate external analysis in this paper.

C. Implications of the results

The system therefore provides the PLP administration output in the form of generated charts and forecast tables so that the programmatic institutional planning is based on actual graduate data. For graduate output consumers, the system offers a direct view of program outcomes and translates them into personalized matches with



possible careers. However, the system output is 100%-based on the input from formative assessment and students' self-report since the survey is the only source of graduates' reflection on their PLP experience.

D. Limitations of the study

The system outputs can be influenced by involuntary and voluntary tracer response bias. The assessment workflow interrupts and prevents the submission of the prediction form in case the mandatory registrar, e.g. the user, has not indicated GPA or OJT grade in the profile. The behavior of the analytical modules is limited to the active server-side Python dependency. The output pie chart in the program overview rendered by `get_data.php` plots upon a static 50/50 profession split instead of actual empirical tracer counts. The landing page statistics are hardcoded and not extracted from the database.

E. Programming language/DBMS used

The application environment is developed with PHP serving for the server side logic, JavaScript is acting as a client to render and Python is executed for machine learning. The database layer is provided by MySQL which is connected through PHP `mysqli` extension and is administered by the `plp_tracer` schema.

F. Recommendations for future research

For future iterations of the system, an update to the analytics dashboard should be implemented to calculate the empirical career distributions as a percentage of aggregated actual `recommended_profession` outputs or submitted employment titles. The forecasting module should also be updated so that the highest 10-5-1 error tables can be exported by the administrators in addition to viewing them as part of the visual representation. In addition, detailed formal

deployment instructions should be written for transferring the system architecture from the current local XAMPP environment to a fully functional web server.

V. CONCLUSION

The PLP Alumni Tracer joins a centralized data collection portal with predictive employment forecasting and Random Forest career classification to produce administrative and alumni-facing analytics from captured assessments, but the utility of the current iteration is virtually limited by tracer data volume, Python server dependencies, as well as the current design decision to "hard-code" certain dashboard metrics as display-only visual placeholders. Future work on the College of Computer Studies Alcalde Jose St. Kapasigan Pasig City, Philippines 1600 system should concentrate on building a more considerable cohort history capable of supporting complex time-series models, conducting the formal accuracy evaluation of the generated forecasts, and further adjusting the database schema to meet national tracer standards.

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